Scouting America

Cybersecurity Merit Badge Workbook

SCOUTMASTER BUCKY

Scouts participating in a Scoutmaster Bucky merit badge opportunity, whether online or in person, should consider using the Cybersecurity merit badge pamphlet for discovery and knowledge, along with the class preparation pages for clarifications, insights, and expectations.

https://scoutmasterbucky.com/merit-badges/cybersecurity/cybersecurity-pamphlet.pdf

https://scoutmasterbucky.com/merit-badges/cybersecurity/cybersecurity-cpp.pdf

REQUIREMENTS 8b and 8c REQUIRE COUNSELOR APPROVAL.

REQUIREMENTS 1a, 4e2, and 5c REQUIRE PARENT / GUARDIAN PERMISSION.

REQUIREMENT 1a: View the Personal Safety Awareness "Digital Safety" video (with your parent or guardian's permission).

PARENT / GUARDIAN PERMISSION: IS REQUIRED.

Parent's / Guardian's Name

Phone or Email

Parent's / Guardian's Signature

Date ☐ *permission*

https://www.scouting.org/training/youth/scouts-bsa/

choose the Digital Safety option

Notes:

REQUIREMENT 1b:	Explain to your counselor how to protect your digital footprint, such as while using social media, mobile device apps, and online gaming.
Social Media:	
Mobile Device Apps:	
Online Gaming:	

REQUIREMENT 1b:	Show how to set privacy settings to protect your personal information, including photos of yourself or your location.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

Notes:

REQUIREMENT 1c:	Discuss first aid and prevention for potential injuries, such as eye strain, repetitive injuries, and handling electronics devices, that could occur during repeated use.

EYE STRAIN:

Prevention:

REPETITIVE INJURIES:

Prevention:

HANDLING ELECTRONIC DEVICES:
Prevention:

REQUIREMENT 1c:	Discuss how to keep yourself physically safe while using a mobile device (for example while walking or biking).
Notes:	

REQUIREMENT 2a:	Relate three points of the Scout law to things people do on the internet or with computers, phones, and other connected electronic devices.
POINT #1:	
Scout Law Point:	
Notes:	

POINT #2:
Scout Law Point:
Notes:
POINT #3:
Scout Law Point:
Notes:

REQUIREMENT 2b:	Discuss with your counselor examples of ethical and unethical behavior in cyberspace. Include how to act responsibly when you encounter situations such as: coming across an unattended or unlocked computer or mobile device; observing someone type their password or seeing it written down near a computer; or discovering a website that is not properly secured.
coming across an unattended or unlocked computer or mobile device:	
observing someone type their password or seeing it written down near a computer:	
discovering a website that is not properly secured:	
REQUIREMENT 2b:	Explain why these situations require good judgement, and how the Scout Law and personal values should guide your actions.
Notes:	

REQUIREMENT 3a:	Describe three types of computer systems that need protecting and explain why with your counselor.
SYSTEM #1:	
Notes:	
Notes:	
SYSTEM #2:	
Notes:	
Notes:	
SYSTEM #3:	
Notes:	
Notes:	

REQUIREMENT 3b:	Explain the "C.I.A. Triad" - Confidentiality, Integrity, and Availability - and why these three principles are fundamental to cybersecurity and discuss with your counselor.
Notes:	

REQUIREMENT 4a:	Define the terms vulnerability, threat, and exploit, and give an example of each that might apply to a website or software product you use and discuss with your counselor.
VULNERABILITY:	
Definition:	
Example:	

THREAT:
Definition:
Example:
EXPLOIT:
Definition:
Example:

REQUIREMENT 4b:	Pick one type of malware (such as virus, worm, Trojan, backdoor, spyware, or ransomware) and find out how it works. Explain what it does and the harm it can cause and discuss with your counselor.
Malware:	
How it works:	
What it does:	
Harm it can cause:	

REQUIREMENT 4c:	Identify two risks of using public Wi-Fi and describe how to reduce or avoid those risks and discuss with your counselor.
RISK #1:	
Risk:	
Notes:	
RISK #2:	
Risk:	
Notes:	

REQUIREMENT 4d:	Describe what spoofing and phishing are.
Spoofing	Phishing

REQUIREMENT 4d:	Describe how to recognize a message or website that might be trying to trick you.
Notes:	

REQUIREMENT 4d:	Explain what steps you should take to protect yourself and others if you come across a message or website that might be trying to trick you and discuss with your counselor.
Notes:	

DO ONE OF THE FOLLOWING (4E1 or 4E2) FOR REQUIREMENT 4E

REQUIREMENT 4e1:	Read an article or a news report about a recent cybersecurity incident, such as a data breach or malware infection. Explain how the incident happened (to the best of your ability based on the information available) and what the consequences are or might be to the victim and discuss with your counselor.
Article Title:	
Publication and Date:	
Summary:	
How the incident happened:	
Consequences for the Victim:	

REQUIREMENT 4e2:	With your parent or guardian's permission, watch a movie or read a book in which cybersecurity plays a significant role. Discuss how cybersecurity topics were depicted and how realistic you think it was.

PARENT / GUARDIAN PERMISSION: IS REQUIRED.

Parent's / Guardian's Name _____ Phone or Email _____

Parent's / Guardian's Signature _____ Date _____ ☐ *permission*

Movie or Book Title:

Summary:

How Cybersecurity Topics were depicted:

How realistic Cybersecurity Topics were depicted:

REQUIREMENT 4f:	Create a list of what is part of your cyber attack surface including all the ways someone could try to access your personal information or devices—such as online accounts, apps, or home networks and discuss with your counselor.
Notes:	
REQUIREMENT 5a:	Describe three technologies that are used to defend a computer or network, such as access controls, antivirus software, firewall, intrusion detection/prevention systems, and Virtual Private Network.
Technology #1:	
Technology #2:	
Technology #3:	

REQUIREMENT 5b1:	Explain to your counselor the importance of installing the latest updates on your computer, why they are needed, and what kinds of problems they can prevent.

Notes:

REQUIREMENT 5b2:	Demonstrate to your counselor how to check for, download, and install the latest updates for your computer or mobile device, or another computer you have permission to use (if you are unable to do this on your computer, you may use an online guide with screenshots to demonstrate this).

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

REQUIREMENT 5b2:	Show your counselor how to verify that your computer or mobile device is up-to-date.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

DO THREE OF THE FOLLOWING (5C1, 5C2, 5C3, 5C4, 5C5, 5C6, 5C7, 5C8, or 5C9) FOR REQUIREMENT 5C

REQUIREMENT 5c: With your parent or guardian's permission, do THREE of the following using a computer or mobile device, and discuss with your counselor:

PARENT/GUARDIAN PERMISSION: IS REQUIRED.

Parent's / Guardian's Name

Phone or Email

Parent's / Guardian's Signature

Date ☐ *permission*

REQUIREMENT 5c1: Describe what makes a good password and why. Set or change an account password to one that is "strong."

Notes:

REQUIREMENT 5c2: Describe multi-factor authentication (MFA) and how it can be used to improve security. Demonstrate how to use an authenticator app or other MFA function.

Notes:

REQUIREMENT 5c3:	Install and set up a password manager. Demonstrate how it works to your counselor.
Notes:	
REQUIREMENT 5c4:	Run a virus scan. Show the results to your counselor.
Notes:	
REQUIREMENT 5c5:	Using a command line or other built-in tool, see what programs or processes are running on your computer. Discuss with your counselor what you see and what surprises you.
Notes:	

REQUIREMENT 5c6:	Use a command line interface to view your computer's open network connections. Discuss the results with your counselor.

Notes:

REQUIREMENT 5c7:	Demonstrate how to back up your data from a mobile device to a local computer or the cloud.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

REQUIREMENT 5c8:	Research best practices for protecting a home computer or network, and make a checklist of FIVE things you and your family can do to stay secure.

Notes:

REQUIREMENT 5c9:	Identify one or more other vulnerabilities on your home computer or network or another computer or network you have permission to use and discuss with your counselor. With permission from the system owner, take the necessary actions to fix it.
Notes:	

REQUIREMENT 6a:	Research and explain to your counselor three situations where encryption is used in cybersecurity. For each situation, describe what kind of encryption is used and why it is important.
ENCRYPTION SITUATION #1:	
Situation:	
Kind of encryption used:	
Importance:	

ENCRYPTION SITUATION #2:
Situation:
Kind of encryption used:
Importance:

ENCRYPTION SITUATION #3:
Situation:
Kind of encryption used:
Importance:

REQUIREMENT 6b:	Show how you can know if your connection to a website is encrypted.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

DO ONE OF THE FOLLOWING (6C1, 6C2, 6C3, or 6C4) FOR REQUIREMENT 6C

REQUIREMENT 6c1:	Create your own encryption code, such as a substitution cipher or code book, and demonstrate using it to encrypt and decrypt a message. Describe the strengths and weaknesses of your code.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

Notes:

REQUIREMENT 6c2:	Download and set up an app (from an official app store) that uses end-to-end encryption. Explain to your counselor what this means, how it works, and why it is more secure than other forms of communication (e.g., SMS).

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

Notes:

REQUIREMENT 6c3:	Use a hashing tool (for example, using SHA or MD5) to create a checksum for a file, document, or piece of text. Have a fellow Scout or your counselor make a change to it, then recreate the checksum and compare the new checksum to the original as a demonstration of file integrity checking.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

Notes:

REQUIREMENT 6c4:	Create your own PGP (pretty good privacy) email key. Share your public key with others (and your counselor). Also, get their public keys and add them to your computer's key ring. Send a message that has been digitally encrypted.

This requirement must be reviewed with your merit badge counselor.

BE PREPARED!

Notes:

REQUIREMENT 7:	Describe to your counselor four electronic devices you encounter that could be connected to the internet, why this connectivity can be useful, what risks are posed by the connectivity, and how they could be protected.
ELECTRONIC DEVICE #1:	
Electronic Device:	
Why this connectivity can be useful:	
What risks are posed by the connectivity:	
How this electronic device could be protected:	

ELECTRONIC DEVICE #2:
Electronic Device:
Why this connectivity can be useful:
What risks are posed by the connectivity:
How this electronic device could be protected:

ELECTRONIC DEVICE #3:
Electronic Device:
Why this connectivity can be useful:
What risks are posed by the connectivity:
How this electronic device could be protected:

DO ONE OF THE FOLLOWING (8A, 8B, or 8C) FOR REQUIREMENT 8

REQUIREMENT 8a:	Learn about a cybersecurity competition, camp, or other activity you could participate in (either now or in the future). Share what you learned with your counselor, including the type of activity, time commitment, and age of participants.

Notes:

REQUIREMENT 8b:	Participate in a cybersecurity competition with members of your troop, school, or some other group approved by your counselor. Either design your own competition or use an existing platform.

COUNSELOR APPROVAL: IS REQUIRED.

Counselor's Name _____ Phone or Email _____

Counselor's Signature _____ Date _____ ☐ *approved*

Notes:

REQUIREMENT 8c:	Give a presentation to your patrol, troop, or another group approved by your counselor, on a cybersecurity topic of your choice. Your presentation must include at least one demonstration and/or hands-on activity.

COUNSELOR APPROVAL: IS REQUIRED.

Counselor's Name _____ Phone or Email _____

Counselor's Signature _____ Date _____ ☐ *approved*

Demonstration or Hands-On Activity:

Notes:

DO ONE OF THE FOLLOWING (9A or 9B) FOR REQUIREMENT 9

REQUIREMENT 9a:	Identify three career opportunities that would use skills and knowledge in cybersecurity.
Career Opportunity #1:	
Career Opportunity #2:	
Career Opportunity #3:	
REQUIREMENT 9a:	Pick one and research the training, education, certification requirements, experience, and expenses associated with entering the field.
Selected Career Opportunity:	
Training Requirements:	
Education Requirements:	
Certification Requirements:	

Experience Requirements:
Expenses Associated with entering this career:
REQUIREMENT 9a: Research the prospects for employment, starting salary, advancement opportunities and career goals associated with this career.
Prospects for Employment:
Starting Salary:
Advancement Opportunities:

Career Goals:
REQUIREMENT 9a: Discuss what you learned with your counselor and whether you might be interested in this career.
Notes:

REQUIREMENT 9b:	Visit a business or organization that does work in cybersecurity. Find out about different work roles, what they do, and how they acquired their knowledge through college degrees or certifications. Share what you learned with your counselor.

Notes: